

Department of Computer Science



Submitted in part fulfilment for the degree of
BEng in Computer Science.

Deep Multitask Learning for News Content Disagreement Detection and Emotion Recognition

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Version 1.0, 2025-April-27

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To everyone who helped me along the way

Acknowledgements

The Viking cluster was used during this project, which is a high performance compute facility provided by the University of York. I am grateful for computational support from the University of York, IT Services and the Research IT team.

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Executive Summary

This research project aims to tackle the problem of online information disorder by exploring the potential of Deep Multitask Learning (MTL) to improve automated news analysis. The main objective of this study is to develop and evaluate an advanced AI model capable of simultaneously detecting factual discrepancies in news content and identifying the emotions expressed within it. The capability to multitask and learn from two different features simultaneously leads to a more robust system that contributes to more effective use against harmful online content by identifying various forms of mis- and disinformation. The central hypothesis was that since misinformation often exploits emotional responses to gain more traction and rapidly spread, a model that could analyse both factual consistencies and affective tone conveyed by the emotions could achieve a deeper contextual understanding and prove more adept at identifying manipulative or false content. The motivation for this arises from the significant societal threat posed by the proliferation of fake news, the limitations of slow manual fact-checking, and the challenges faced by existing automated methods; despite having advanced models like RoBERTa, they still struggle with subtle manipulation and also require large amounts of labelled data.

The study followed a structured Waterfall methodology, which led to designing and implementing a multitask neural network with RoBERTa-base transformer architecture. The Roberta transformer was chosen for its superior performance in capturing linguistic nuances compared to other models like the older BERT and efficiency-focused models like ALBERT and Distilbert. The architecture of the model featured a shared RoBERTa encoder that generates contextual text that is then fed into the two separate task-specific prediction heads: one for news disagreement classification (Misinformation checker that outputs True or False) and another for multiple category emotion recognition across 28 different emotions. Several public datasets were used to train such a model, including Politifact, Hyperpartisan News, FNC, ByteDance and GoEmotions, which aimed to cover different facets of disagreement and emotion to allow the model to understand richer nuances in misinformation. To implement all these datasets, preprocessing was a crucial step, which involved unifying the diverse labels in the disagreement datasets into a binary format so that they could be trained together and additionally implementing class weights to mitigate data imbalances, which would cause the model to be biased and inaccurate. The model

was trained using PyTorch, the Adamw optimiser, and task-specific loss functions(BCEWithLogitsLoss for disagreement and Cross-entropy loss for emotion), gradient clipping and saving the best model based on the average validation loss across tasks. The performance evaluation of the model involved testing with standard metrics like F1-score, accuracy, and the AUC(Area Under the Curve) on a ROC(Receiver Operating Characteristic) Curve, which was compared against a multitask LSTM and single-task RoBERTa model baselines.

The results of the performance evaluation provided a mixed picture regarding the efficacy of the MTL approach. For disagreement detection, the multitask RoBERTa clearly outperformed the LSTM baseline, particularly on the complex FNC dataset, therefore validating the choice of transformer architectures' strength. However, compared to a single-task RoBERTa model, the multitask version performed worse on the BD dataset (F1-score: 0.78 vs 0.87) but marginally better on the challenging FNC data(F1-score: 0.45 vs 0.43). The overall low performance on FNC across all models highlighted persistent difficulties within automated systems with understanding a more complex, specific task or dataset. Although in contrast to the overall decent performance for disagreement detection, the emotion recognition task suffered significantly with the MTL approach. The multitask RoBERTa model slightly improved upon the multitask LSTM model(F1-score: 0.31 vs. 0.29) and was substantially outperformed by the single-task RoBERTa model that focused solely on emotion(F1-score: 0.31 vs. 0.41). Indicating a considerable negative inference from the disagreement task, which is hindering the model's ability to learn emotional nuances effectively. The potential causes for this problem may include task interference during optimisation, differing convergence rates, or the disagreement task dominating the shared encoders' representation learning. As well as the chosen strategy of optimising for average validation loss might have also contributed to favouring overall balance at the expense of peak individual task performance. In conclusion, this research successfully implemented an advanced multitask model and confirmed the effectiveness of RoBERTa transformers. The study emphasised the major challenges of multitask learning, particularly in balancing the objectives of different tasks and avoiding negative interference between them. Future research should prioritise the development of advanced multitask learning techniques and integrate diverse data types to effectively address the complexities of misinformation in the online landscape of information disorder.

Statement of Ethics

This project develops an AI model for detecting news disagreements and recognising emotions. It utilises publicly available datasets, including Liar(Politifact), Hyperpartisan News, GoEmotions, FNC-1, and WSDM Fake News Classification(BD), and adheres to their respective licenses and terms of use for non-commercial academic research. Consent for data usage is based on the public nature of the source materials, such as public statements, news articles, and Reddit posts.

An ethical considerations checklist confirmed that the project does not involve human participants, restricted datasets, work in high-risk countries, or human/animal materials. Therefore, there are no significant ethical implications requiring formal ethical approval.

1 Introduction

The rapid expansion of digital communication has enabled widespread information sharing across the world; however, with its growth comes a proliferation of misinformation, disinformation and fake news, which has the potential to negatively impact society, individuals, democracy and journalism [1]. Studies show that the scale of this issue is substantial and that fake news stories can generate more engagement and shares than factual reports from well-known news sources [1]. This phenomenon links to the broader issue of "information disorder" that has emerged in the digital era and pollutes the online environment [2].

To combat this information disorder, traditional methods exist, such as manual fact-checking by Politifact and Snopes. Although these organisations still face significant challenges because of the sheer volume and speed at which false information can spread online, they cannot keep pace due to the time-consuming nature of manual fact-checking [1]. Therefore, research into automated detection methods has increased, particularly supervised machine learning and deep learning techniques [3]. Early approaches used traditional linguistic features, which had major limitations; however, as research has continued, the addition of advanced models like pre-trained transformers (e.g., BERT[4], RoBERTa[5]) has significantly improved the ability of automated detection methods to understand text contextually. Nonetheless, challenges remain due to these methods being supervised; they require large labelled datasets, data heterogeneity across different sources, and the difficult nature of detecting subtle forms of misinformation that are misleading or out-of-context information [1]. Some of the automated methods that exist include approaches like fact verification [6], stance detection [7], and multimodal analysis [8][9].

This project explores an advanced technique called Deep Multitask Learning (MTL) specifically for news content analysis [10][11]. The MTL technique works by training a single model on multiple related tasks simultaneously and, in doing so, potentially improving the generalisation of the model with richer learnt representations. The central hypothesis for this study is that analysing news content not just for factual disagreement but also for emotions conveyed can lead to more robust detection of misinformation [10]. The motivation for this idea is that misinformation often leverages strong emotional responses to increase its spread online [10]. Thus, by

incorporating emotion recognition as a complementary task, the model may have an enhanced ability to understand context and identify manipulative content [10].

This study proposes and evaluates an MTL model built utilising the RoBERTa transformer architecture [5]. The model features a shared encoder with two task-specific prediction heads: one for binary news disagreement detection and another for multi-class emotion recognition. The aim of this research is to develop a model capable of identifying various mis- and disinformation types by simultaneously learning from factual inconsistencies and emotional expression, contributing to more effective automated tools against information disorder [2]. The methodology, implementation details, and evaluation of this MTL approach are presented in the subsequent chapters.

2 Literature Review

2.1 Background

The exponential growth of social media and web communication has given rise to the ability for anyone to create content and share information online. With this comes the spread of misinformation or fake news. Fake news spreads unreliable, harmful and malicious content that can threaten democracy, journalism and freedom of expression. Fake news stories can even generate much more interest on social media than major news websites. In 2016, during the U.S. presidential elections, the top 20 fake election stories generated 8,711,000 shares, reactions and comments on Facebook. In contrast, the top 20 most-discussed election stories by 19 major news websites only garnered a lower 7,367,000 [1]. The proliferate spread of this fake news can have major impacts on society. This information disorder built on disinformation, misinformation and malinformation pollutes the web and instead of an increase in web access globally connecting people, it is dividing us [2]. Although the majority of online content is not news-related and thus isn't as high a risk of creating disinformation, there is still a notable amount of the web that contains vital information that needs to be trustworthy and reliable. Consequently, this has led to the creation of satirical news sources like "The Onion" and fact-checking websites such as "PolitiFact" and "Snopes" as forms of detecting fake news. This form of fake news detection, even though effective, comes with drawbacks for instance, they all require human expertise to manually assess the credibility of a claim, which is very time-consuming, and in addition to that, fact-checking websites often constrain themselves to a particular domain of interest, which leaves a gap in the credibility of more generalised content. Because of these drawbacks, research has been done on using supervised machine learning or neural networks to detect fake news and combat information disorder. The problem these models face is determining whether a news item is true or false, which can be traditionally known as a supervised binary classification problem [3], [6], [8].

2.2 Supervised Learning Approaches for Detection

Supervised learning approaches have become the basis for research into automated fake news or misinformation detection. Key tasks like fact verification, stance detection and disagreement detection are often seen as supervised classification problems, where models learn to map input text to labels [3][7]. Initially, this approach relied on carefully engineered traditional linguistic features, which lacked significant performance; however, with the addition of deep learning techniques, particularly the adoption of Transformer-based models like BERT and its variants, such as RoBERTa, this approach has seen significant performance improvements. Due to these models excelling at capturing deep contextual understanding of the text, surpassing previous methods[12].

Despite these advancements, supervised learning approaches still face challenges. The main bottleneck with this approach is that it relies heavily on large volumes of labelled training data. Creating those datasets is a time-consuming and expensive process, which means the scale and diversity of resources will be limited. Speaking of labelled data, datasets often employ different annotation schemes ranging from a truthfulness rating [12] to stance categories [7] or simple binary classification. This leads to data heterogeneity, which becomes an issue when combining multiple datasets. Thus, the solution is to preprocess the labels of each dataset required for the specific task to have the same label annotation scheme.

Moreover, even with the addition of accurately labelled data, the challenge of subtlety and context still comes. Usually, accurately labelled data is all a supervised learning model needs to achieve adequate performance or accuracy, but with the specific task of classifying misinformation, that may not be enough. This is because misinformation is not always characterised by outright false statements but rather by misleading framing, out-of-context information or subtle manipulation. Even with transformer models that offer improved contextual understanding, detecting such nuances in misinformation is still tricky. That is why there are research attempts to address this weakness. One example of this is evidence-aware models that explicitly cross-reference claims against external information to aid in verification [6], or an additional example is multi-modal models that combine text and image or video to classify misinformation due to misinformation often being associated with an image or video in modern news [8][9]. The research into these models aims to identify ways to uncover more significant nuances in the data and thereby gain a deeper contextual understanding to classify misinformation.

Finally, the last challenge that supervised learning faces is class imbalances. Class imbalance in the context of news disagreement refers to a dataset with more true news stories than false news stories or vice versa. If left unchanged, the model would be biased towards predicting one classification over the other. However, there are many ways to solve this issue, like undersampling, where all classes have the same number of entries as the minority class or oversampling, where all classes are duplicated until they have the same number of entries as the majority class and class weighting during training, which gives more importance in the model to the minority class. All these methods help to mitigate bias and eliminate class imbalance in the dataset.

2.3 Pre-trained Language Models

Expanding on transformer models, the advent of large pre-trained language models based on transformer architecture has revolutionised natural language processing(NLP). These models are pre-trained on vast text datasets to learn rich linguistic representations, effective for news disagreement and emotion classification. BERT was a foundational bidirectional model developed by Google that utilises the Masked Language Model(MLM) and Next Sentence Prediction (NSP) pre-training objectives [4].

Improving on BERT came RoBERTa, which was developed by Facebook AI [5]. This model evaluated BERT and implemented several modifications to significantly improve the performance of tasks by optimising training and adding more data (approximately 160 gigabytes of text) compared to BERT. It also employs a dynamic masking approach, which changes the masking pattern applied to the input on each epoch. It trains with larger batch sizes with more extended sequences and uses a more extensive encoding vocabulary for better handling morphology. These modifications result in more robust language representations and better NLP accuracy than the original BERT model.

Other variants of BERT also exist but mainly focus on efficiency over performance. Distilbert uses knowledge distillation to create a smaller and faster version of BERT while still retaining adequate performance [13]. Similarly, ALBERT implements parameter reduction techniques like cross-layer parameter sharing and Sentence Order Prediction(SOP) to accomplish its better efficiency compared to BERT [14].

2.4 The Role of Affective Features: Emotion Recognition

Beyond the literal content of the data and the objective accuracy of its information, research suggests that taking note of the emotional dimension within the text plays an important role in the ecosystem of information disorder [10]. News content that elicits strong negative emotions, such as anger and fear, or positive emotions, like joy and admiration, tend to gain more attention than neutral news articles that do not lean toward eliciting an emotion. As a result, misinformation or biased news articles can spread more rapidly across social networks. Therefore, recognising the emotional dimensions of news has become an essential step in combating misinformation.

This is the motivation behind Emotion Recognition being a complementary task to News Disagreement Detection. Furthermore, training models to identify emotions conveyed in text addresses the challenge of subtlety and context by providing additional information that could improve the models' contextual understanding and find more nuances in the data. An excellent dataset for this is GoEmotions, developed by Google, which has 28 categories of emotion and labelled text data [15]. Identifying manipulative or inflammatory language or recognising patterns of emotional expression can offer valuable clues that directly complement the identification of misleading content or misinformation in news disagreement detection.

2.5 Leveraging Social Context and Network Dynamics

Purely content-based analysis based on articles being true or false has limitations. That's where the research into leveraging social context and network dynamics comes in. The key notion of this method is that information spreads, and the people who spread it or the reaction garnered from others provide strong indicators of the nature of the content.

This method incorporates features derived from the surrounding social network structures and propagation dynamics. This method is an alternative approach to misinformation detection. The techniques in this method include analysing share and retweet patterns, the depth and breadth of information cascades [16][17], the structure of user interaction within the networks [18], and metrics related to source and user credibility [11]. These approaches often use a semi-supervised learning method that uses the

graph structure of social networks or a large amount of unlabelled user interaction data [19]. This can partially address the data scarcity issue that supervised models face.

2.6 Unsupervised Detection Methods

If data scarcity is a huge challenge to address, then the best alternative to supervised learning would be unsupervised learning. This approach aims to detect misinformation without reliance on labelled data.

These methods seek inherent patterns within the unlabelled data that can distinguish misinformation from factual content. The techniques used in this method involve clustering articles or sources based on writing style or topic consistency, detecting anomalies in propagation patterns or analysing the consistency of information across multiple conflicting articles, which utilises principles from 'truth discovery' methodologies [20][21].

2.7 Advanced Techniques: Multitask Learning for Enhanced Detection

Lastly, in order to build a model that is more holistic, robust and efficient, an advanced technique like Multitask Learning (MTL) is utilised. MTL is a technique used to improve generalisation and capture richer representations by training a single model to perform multiple tasks simultaneously. This addresses challenges like subtlety, heterogeneity, and model efficiency.

An MTL architecture related to this topic typically involves a shared base encoder, such as RoBERTa, that learns linguistic features, with task-specific heads that branch off to make predictions for each task [11][10]. Using this approach allows the model to learn shared knowledge across tasks, which leads to cross-task enhancement where learning one task improves performance on another and acts as an implicit form of regularisation that improves generalisation.

Nonetheless, with MTL's benefits, other challenges remain. Due to the ever-evolving fake news tactics, the standard MTL may need to incorporate alternative techniques like adversarial training to improve robustness against unseen manipulated data [9]. Moreover, misinformation is a global problem that can transcend language barriers; therefore, research that explores cross-lingual transfer learning is emerging to enable models trained

2 Literature Review

primarily on one language to adapt and function in other languages [22]. These techniques show a glimpse of the continuous and future research and improvements made to neural networks to tackle the ever-growing spread of misinformation.

3 Methodology

This chapter covers the methodology used, understanding information disorder, defining objectives for news disagreement detection and emotion recognition, data acquisition and feature engineering for Multitask Learning, and neural network architecture for disagreement detection and emotion recognition.

The methodological approach for this project was the Waterfall method[23]. The core principles of the Waterfall method are that all requirements and planning must be done before the project begins and that the next phase does not start until the previous phase has been completed. This approach is appropriate due to the project's clear objectives. Since the tasks and expected outcomes are relatively straightforward and lack variability, gathering requirements and planning before beginning the project will help streamline the design and implementation phase. It also creates a structured plan to follow sequentially, improving workflow. Additionally, the Waterfall method uses five categories: Requirements, Design, Implementation, Verification, and Deployment and Maintenance. Given the project's proof-of-concept nature, the final Deployment and Maintenance category is not applicable. Each step of this project can be seen in Figure 3.1: Waterfall Methodological Phases.

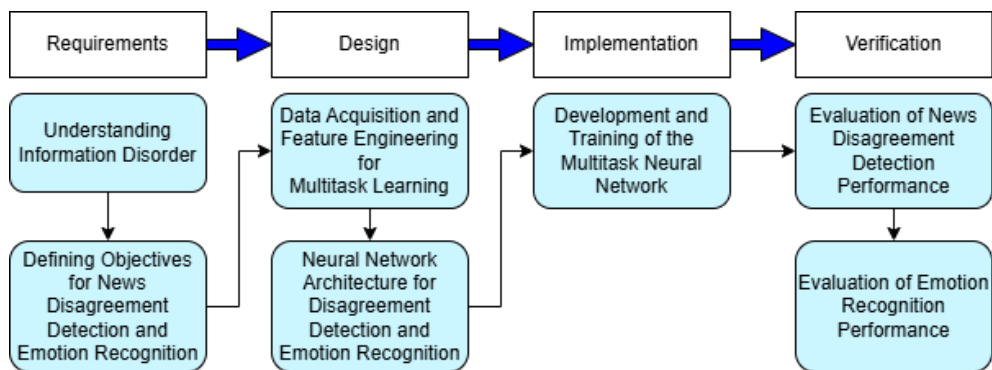


Figure 3.1: Waterfall Methodological Phases

3.1 Requirements Phase

3.1.1 Understanding Information Disorder

The literature review shows various methods and designs for building a multi-task neural network that specialises in news disagreement detection. To find the appropriate objectives, techniques, and designs for this project, understanding information disorder is imperative to achieving the goals of this neural network. Information disorder can be defined as the wide range of ways information can be spread incorrectly, misleadingly or harmfully in the modern communication landscape. Information disorder can be divided into three main categories: Disinformation, Misinformation and Malinformation. [2]

Type	Description	Falseness	Intent to Harm	Information Category	Notes
Satire or Parody	Content meant to humorously critique, but can fool individuals if taken literally.	Implicitly False / Exaggerated	No (usually)	Misinformation	Intent is humor/critique, but potential to mislead if context is missed.
False Connection	Headlines, visuals, or captions that don't accurately reflect the main content.	False (in the implied connection)	Potential / Often Yes	Mis/Disinformation	Can be accidental (Mis) or deliberate clickbait/framing (Dis).
Misleading Content	Using information selectively or twisting it to frame an issue or person falsely.	False (in the overall framing)	Yes	Disinformation	Deceptive use of (sometimes true) information to create a false narrative.
False Context	Sharing genuine content but with fabricated contextual information.	False (in the added context)	Yes	Disinformation	The original content is true, but the surrounding story is false.
Imposter Content	Falsely claiming to be from or acting as a genuine source (impersonation).	Very False (source is false, content often too)	Yes	Disinformation	High intent to deceive by leveraging the trust in the impersonated source.
Manipulated Content	Altering genuine information or imagery to deceive (e.g., "photoshopping").	False (content is altered)	Yes	Disinformation	Deliberate alteration makes the content false with the intent to mislead.
Fabricated Content	100% false content, created solely to deceive and potentially cause harm.	Entirely False	Yes	Disinformation	Purely designed to mislead and often cause harm.

Figure 3.2: The Seven Types of Mis- and Disinformation

As presented in Figure 3.2, there are seven different types of Mis- and Disinformation defined by Claire Wardle[2]. The figure does not contain the Malinformation category since it is described as true information that is spread with the intent to harm, and this project only focuses on false information. Mis- and Disinformation are detailed as false content but differ in intent. Misinformation is when false content is shared mistakenly or without intent to harm, whereas Disinformation is false content with the intent to harm. This project will focus on both categories for news disagreement detection, due to intent not being a criterion for the tasks of this neural network. The seven types of Mis- and Disinformation are Satire and Parody, False Connections, Misleading Content, False Context, Imposter Content, Manipulated Content, and Fabricated Content. These topics are at the core of information disorder. Therefore, the goal of this multi-task neural network is to be able to detect these types of Mis- and Disinformation with appropriate accuracy.

3.1.2 Defining Objectives for News Disagreement Detection and Emotion Recognition

The motivation of the news disagreement detection task shall be to address the spectrum of information disorder by attempting to identify the seven types of Mis- and Disinformation. Therefore, the objective of this news disagreement detection shall be to perform binary classification on text to identify characteristics associated with Mis- and Disinformation, with the output being whether the text is true or false.

Additionally, the objective of emotion recognition shall be to classify the text input into a perceived emotion category, thus outputting the emotion that is most likely conveyed through the given text.

Following these simple objectives, the prospective requirements gathered for this project are represented in Table 3.1 below. The table is organised by category, in the order of the Waterfall method, and has a necessity column to show the importance of each requirement.

Table 3.1: Prospective Methodology Requirements with Necessity Levels

ID	Category	Necessity	Requirement Description
R1.1	Architecture	Must Have	The implementation shall be designed as a multi-task neural network, based on a pretrained Transformer architecture.

Continued on next page

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ID	Category	Necessity	Requirement Description
R1.2	Architecture	Must Have	The architecture will have two distinct output heads: one for binary classification on disagreement detection and one for emotion classification.
R1.3	Functionality	Must Have	The model shall be capable of accepting news-related text as input.
R1.4	Functionality	Must Have	The disagreement detection shall produce a binary classification (True/False) with the associated probability score.
R1.5	Functionality	Must Have	The emotion recognition feature shall classify input text into one of the emotion categories.
R1.6	Functionality	Should Have	The implementation shall provide a method to perform both tasks on new text inputs.
R2.1	Data Handling	Must Have	The implementation shall be designed to train on multiple datasets.
R2.2	Data Handling	Must Have	A data preprocessing function shall be developed to correctly map source labels to the required binary/multi-class formats, due to multiple varying datasets being used.
R2.3	Training	Should Have	The implementation shall use techniques to mitigate bias in classes for the disagreement task.
R2.4	Training	Must Have	The training process will use loss functions that are appropriate to the task they are assigned to.
R3.1	Evaluation	Must Have	The performance of the disagreement detection shall be evaluated using standard binary metrics on a test dataset.
R3.2	Evaluation	Must Have	The performance of emotion recognition shall be evaluated using standard multi-class metrics on a validation dataset.

Continued on next page

ID	Category	Necessity	Requirement Description
R3.3	Evaluation	Should Have	The evaluation process shall include a qualitative assessment of the model's sensitivity to examples of different information disorder types.
R4.1	Implementation	Must Have	The implementation shall use Standard libraries for relevant supporting tasks.

3.2 Design Phase

3.2.1 Data Acquisition and Feature Engineering for Multitask Learning

The design of an effective implementation for news disagreement detection requires careful consideration of data sources and feature representation. As surveyed by Zhou and Zafarani[1], various approaches exist suitable for news disagreement detection. The key to the design of this project will be to choose the most suitable features and learning paradigm from the given objectives and available resources. Several types of features and methods were considered from the literature. These include social context and network features, multi-modal features, evidence-based verification, and content-based representation.

The social context and network features method utilises external content, such as user engagement patterns[19][16], source credibility metrics[11], information propagation[17] and reply-aided interactions[18]. The pros of this type of method are that it can capture how information spreads, identify inauthentic coordinated behaviour, e.g. bots, perceive community reactions, and detect manipulated information. As a result, this method is compelling for understanding real-world impact and dynamics. However, this would require complex, large-scale social network data to gain that understanding, significantly increasing data acquisition challenges and feature engineering complexity. Moreover, this approach is deemed outside the project's scope because it uses social network content instead of news content.

The multi-modal feature method integrates visual data alongside text using a specialised model capable of processing multiple modalities[8][9]. The pros of a multi-modal method are that it enables the detection of visual manipulations, e.g. a doctored image and false connections through inconsistencies between the text and visuals. This method can be very

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useful for current online news platforms because visuals are often included. Nevertheless, with the visuals comes a requirement for large, well-aligned, multi-modal datasets, which are rarer than text-only datasets and require higher computational resources to train and use the model. Due to the scarcity of datasets and higher resources required to use this approach, this project is not as feasible.

The evidence-based verification method explicitly compares textual claims to external knowledge bases, for example, Wikidata or retrieves and analyses evidence documents from a reliable corpus, like Wikipedia[6]. The pros of this method are that it allows for a firm factual grounding for its claims by verifying against external facts. Therefore, it yields a high accuracy for claims that are directly verifiable within the source of knowledge. However, with this comes a heavy reliance on the coverage, accuracy and recency of the external knowledge source. The model's accuracy will be significantly impacted if the knowledge source lacks any of those points. This method requires sophisticated implementation for information retrieval and natural language inference, and may struggle with non-factual disagreement types like biased or opinionated stances. Therefore, this method's heavy reliance on an external knowledge source and its sophisticated components may not be the best fit for this project.

Finally, the content-based representation method uses a pre-trained transformer model to extract signals directly from text. Unlike other methods that incorporate external data, social context or external knowledge bases, this approach uses Natural Language Processing(NLP) techniques to analyse semantics, syntax, style, and contextual information contained purely in the words and sentences of the provided text. This allows the model to learn the disagreement detection and emotion recognition tasks directly from the language used in the data[3][10]. The pros are that it can leverage readily available text data without complex data integration. It allows for deep semantic understanding using a modern NLP model while avoiding potential noise or sparsity associated with the social context and network feature methods. The con of using this method is that it does not use an external context like other methods, so potential context that could further aid classification is ignored, which means performance relies heavily on the underlying pre-trained model's ability to understand language nuances. Overall, this method is the most appropriate for the project's objectives to classify news disagreements and emotions, with readily available data and fewer resources required to implement than other methods.

Following this design approach, the data acquisition may begin because the chosen method would affect the types of data used. The content-based representation approach uses only text to classify tasks. Therefore, gathering an array of data sources that are text-based and news-related would be beneficial for identifying the seven types of Mis- and Disinformation

in news articles. For this project, five text-based datasets have been chosen, four of which will be used in disagreement detection and one for emotion recognition. The reason for using four datasets for disagreement detection, unlike one for emotion recognition, is that the disagreement task has to have more data on different styles of news disagreement in order to identify the seven information disorder types appropriately. The datasets chosen are Politifact('liar')[12], Hypertisan News Detection[24], Fake News Challenge 1(FNC)[7], Byte Dance [25] and GoEmotions [15]. A brief rationale on why each dataset was chosen is provided in Table 3.2.

Table 3.2: Dataset Rationale

Dataset Name	Reason for Selection / Contribution
Politifact ('liar')	Provides benchmark factuality labels, essential for detecting fabricated or manipulated content for the Disagreement task.
Hyperpartisan News Detection	Addresses extreme bias and framing, adding signals beyond pure factuality, relevant to identifying Misleading Content.
FNC Data (FNC-1)	Focuses on headline-body consistency, crucial for identifying False Connection or misleading headlines within the Disagreement task.
BD Data	Evaluates consistency between related claims (headlines), useful for detecting contradictory narratives. (Note: Specific origin undetermined).
GoEmotions	Supports the auxiliary emotion recognition task in the MTL setup; enables exploring emotion/disorder links and potentially improves representations via regularization.

3.2.2 Neural Network Architecture for Disagreement Detection and Emotion Recognition

Based on the comparison presented in Figure 3.3, the chosen pre-trained transformer used is the RoBERTa transformer developed by Facebook AI. RoBERTa is the best choice because of its superior performance, optimisa-

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tion, and pre-training data scale. RoBERTa is a better choice than BERT because it is built directly from BERT and enhances BERT's pre-training by using a larger data scale and an optimised Masked Language Model (MLM) that surpasses the standard MLM used in BERT. This results in more robust language representations and higher accuracy on classification tasks. While DistilBERT and ALBERT are smaller, faster and more efficient than RoBERTa, they do not perform as well when benchmarked. The model's ability to capture as many nuances and subtle linguistic cues as possible is a key part of learning disagreement and emotion detection. Therefore, the superior performance of RoBERTa was a key factor in choosing it.

The core architecture of this project will utilise the RoBERTa pre-trained transformer model as the foundational component, specifically, the roberta-base variant of the model, which has a size of approximately 125 million parameters, 12 layers and 768 hidden dimensions. The architecture will also follow a paradigm focused on Multi-Task Learning (MTL), leading to a single model that can perform both disagreement detection and emotion recognition concurrently. The roberta-base model serves as the shared encoder within the MTL framework. The shared encoder processes input text to generate rich and contextualised embeddings that capture semantic and syntactic information, forming a common base for both tasks. Along with the shared encoder comes two prediction heads that attach to it. Each head is distinct, task-specific, and consists of a feed-forward network. The final layers of the two heads will differ by assigned task; the disagreement head will output one logit for binary classification in its final layer, whereas the Emotion head will output twenty-eight logits, one for every emotion class in the GoEmotions dataset, for multi-class classification. The input for both heads will be the pooled output from the roberta-base encoder. The operational mechanism, which ensures the right input goes to the correct task-specific head, will be a conditional forward pass that looks into the input metadata for task type, either disagreement or emotion and then directs the output of the encoder to the correct task-specific head. Furthermore, the regularisation technique of dropout will be applied to the model to mitigate overfitting and improve model generalisation. In summary, the proposed architecture of this model will have parameter efficiency by sharing the large RoBERTa encoder, therefore reducing the total parameters while also reducing its memory footprint as well as faster training convergence per parameter update and potential cross-task knowledge transfer, enabling the model to learn representations that capture emotional nuances that also inform disagreement detection, or vice versa, thus improving robustness.

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Feature	RoBERTa (roberta-base)	BERT(bert-base)	DistilBERT(distilbert-base)	ALBERT(albert-base-v2)
Developer	Facebook AI	Google AI	Hugging Face	Google AI
Approx. Parameters	~125 Million	~110 Million	~66 Million	~12 Million
Pre-training Data Scale	Very Large (160GB+), diverse sources	Standard (16GB), Books & Wikipedia	N/A (Learned from BERT)	Standard (16GB), Books & Wikipedia
Key Pre-training Task(s)	Optimized Masked LM (MLM) with <i>dynamic</i> masking; No Next Sentence Prediction (NSP) task.	Standard Masked LM (MLM) with <i>static</i> masking; Includes Next Sentence Prediction (NSP) task.	Knowledge Distillation from BERT (using MLM, Cosine, KL divergence losses).	Masked LM (MLM); Sentence Order Prediction (SOP) task (replaces NSP).
Key Design Principle / Architecture	Focus on optimizing BERT's pre-training methodology for robustness and performance.	Foundational bidirectional transformer encoder stack.	Compressing BERT into a smaller, faster model via Knowledge Distillation.	Reducing model size significantly via Parameter Sharing and Factorized Embeddings.
Primary Strength(s)	State-of-the-art performance on many NLU benchmarks; Robust representations.	Strong, well-understood baseline; Good general capabilities.	High computational efficiency (speed/memory); Good balance for less complex tasks.	Excellent parameter efficiency (very small size for its power).
Primary Limitation(s) / Trade-off	Higher computational cost (memory/time) than smaller variants.	Pre-training is less optimal than RoBERTa; the NSP task can hinder performance.	Reduced performance on complex tasks requiring deep nuance compared to larger models.	Can slightly underperform larger RoBERTa/BERT on some challenging benchmarks; Unique training dynamics.

Figure 3.3: Comparison of different Pre-trained Transformer Base Models

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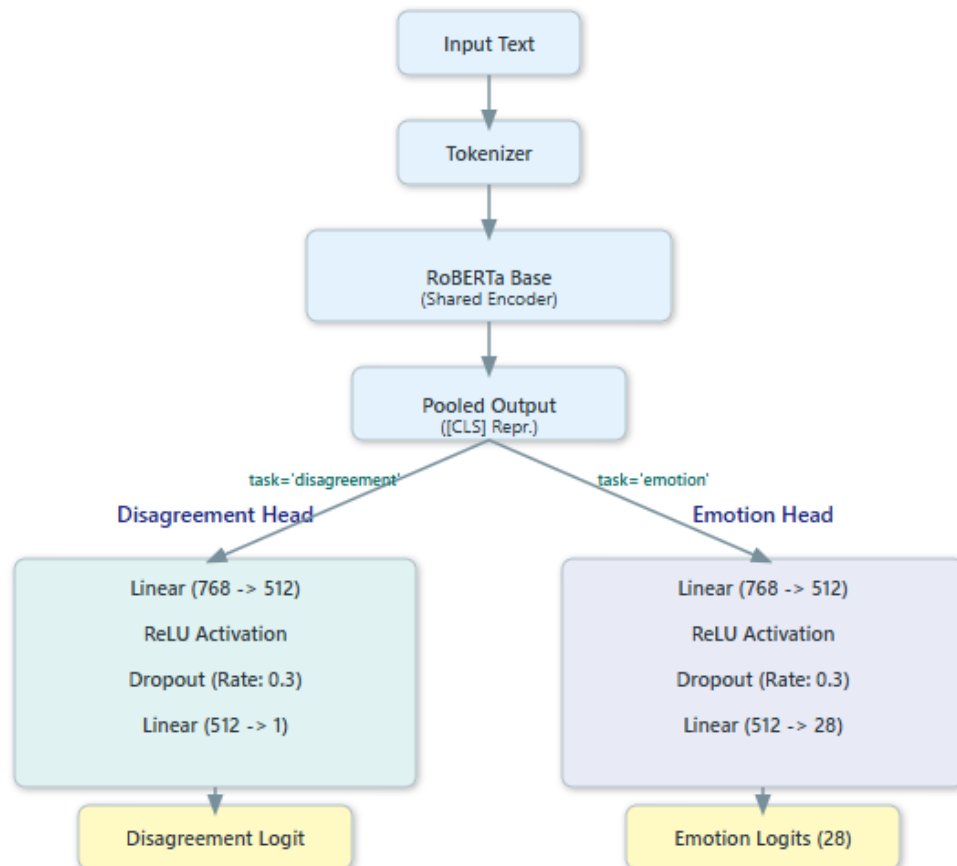


Figure 3.4: Process diagram of model architecture

4 Implementation Phase

This chapter will cover the final implementation used to create the multi-task neural network for news disagreement detection and emotion recognition.

4.1 Development and Training of the Multitask Neural Network

Table 4.1: Python Libraries and Dependencies Utilised

Library	Version*	Primary Role in Implementation
torch	[1.13.1]	Core deep learning framework; model definition, training, tensor operations.
transformers	[4.25.0]	Provides RoBERTa model implementation and tokenizer (Hugging Face).
datasets	[2.10.0]	Loading benchmark datasets (e.g., <code>liar</code> , <code>go_emotions</code>) from Hugging Face Hub.
scikit-learn	[1.2.0]	Evaluation metrics (accuracy, F1, etc.), data splitting, label encoding.
nltk	[3.8.1]	Sentence tokenization (<code>sent_tokenize</code>) for inference processing.
pandas	[1.5.2]	Data loading from CSV files, Data-Frame manipulation during preprocessing.
numpy	[1.23.5]	Numerical operations, array handling (often a dependency of other libraries).
os	[Python Std. Lib.]	Listing directory contents (loading local CSVs).
warnings	[Python Std. Lib.]	Suppressing specific warning messages during execution.
math	[Python Std. Lib.]	Checking for infinite/NaN values (e.g., during weight calculation).

4 Implementation Phase

Table 4.1 lists the libraries and dependencies needed for this project, which is written in Python and will use the PyTorch library as its core deep learning framework.

Building on the model architecture described within the methodology section, we can go into more detail about the task-specific head structure. Each head consists of a Linear layer that projects the 768-dimensional RoBERTa output to an intermediate dimension of 512, followed by a ReLU activation function to provide non-linearity, a Dropout layer of $p = 0.3$ for regularisation to prevent overfitting and a final Linear Layer that projects the task-specific output dimension of either 1 or 28.

Furthermore, a key step in the implementation is preprocessing the datasets into a disagreement or emotion task. Disagreement tasks are handled differently from emotion tasks; therefore, they go through different preprocessing. The disagreement datasets must go through label unification to map the labels into a consistent binary format for binary classification, unlike the emotion task, which gets preprocessed into 27 integers and does not need unification because it only has one dataset. Additionally, in the preprocessing function, there is often a class imbalance when using multiple or unbalanced datasets. To mitigate the imbalance, the preprocessing function counts the number of 0 and 1 labels for the disagreement datasets, finds the imbalance, and then assigns a weight to the dataset class based on the count difference, and for the emotion dataset, it counts the frequency of each emotion category and then assigns weights based on the rarity of the emotion within the dataset, the rarer, the more weight is added and the more common, the less weight added. This weight is then passed to the loss function during training to achieve a weighted loss and handle the class imbalance so that each class has equal importance.

Training the model was based on standard machine learning practices with multi-task adjustments; datasets were split into training, validation and test sets. The randomised sample order was turned on for training and off for validation. Moreover, two loss functions are deployed to calculate the loss for a specific task; the disagreement task uses binary cross-entropy with logits loss, and the emotion task uses cross-entropy loss. The optimiser employed is AdamW, a common choice for transformer models because of its effective weight decay handling. This optimiser also has a learning rate of $2e-5$. Additionally, gradient clipping was applied after the backward pass to improve training stability. Also, to ensure the best model was saved, the best validation loss is checked after every epoch, and if the loss is lower than all previous epochs, the model is saved.

Following the training comes the evaluation section of the implementation. Within validation and testing, the model mode is set to evaluation. After running the validation or test function, metrics are calculated on the model's

4 Implementation Phase

accuracy. These metrics include accuracy, F1-score, Precision, and Recall, which provide a more balanced performance measure.

5 Verification Phase

5.1 Model Rationale

Table 5.1: Dataset Label Mapping and Class Weighting Logic

Dataset Name	Model Label Mapping	Calculated Weight(s)
Politifact ('liar')	Codes 0, 4, 5, -1 → 1 (Disagree) Codes 1, 2, 3 → 0 (Agree)	weight = 1.2835
Hyperpartisan	'true' → 1 (Disagree) 'false' → 0 (Agree)	weight = 1.7101
FNC	'agree' → 0 (Agree) 'disagree'/'discuss'/'unrelated' → 1 (Disagree)	weight = 4.3786 (Train Split)
BD	'agreed' → 0 (Agree) 'disagreed' → 1 (Disagree)	weight = 11.2476 (Train Split)
GoEmotions (simplified)	Use original ID (0-27) directly	Full tensor (shape [28]) using 'balanced' heuristic

The verification phase will focus on the selected multitask RoBERTa-based model. It will evaluate its performance against an LSTM model as a classical deep learning baseline, which will be used to quantify the transformer architecture's performance contribution. Two single-task models will also be used to test whether single-task models perform better than multitask models.

The best epoch was selected from the final implementation and will be used as the final results of the model. Both the final proposed model and the baseline models ran for 10 epochs and saved the model weights of the epoch with the minimum average validation loss over all epochs. Minimising the average loss across tasks or datasets is key to selecting the best model that has good generalisation before significant overfitting.

Additionally, to evaluate the final model accurately, there must be complete consistency in data preprocessing and hyperparameters across the final proposed and baseline models. Therefore, the label mapping and weights in Table 5.1 will be the data preprocessing used on all models. The weight in disagreement tasks is calculated by dividing the label counts of 0 by the label counts of 1 because it is a binary classification problem and then is applied to the BCEWithLogitsLoss function. Similarly, the emotion tasks use a whole class weight tensor applied to a CrossEntropyLoss function.

5.2 Performance Evaluation and Results - News Disagreement Detection

The final performance metrics achieved by the selected multitask RoBERTa model on the two test sets were test_bd (ByteDance): Accuracy(The overall percentage of predictions the model got right): 0.9695, F1-score(The mean of Precision and Recall, balancing both metrics into a single score): 0.7786, Precision(Of the items the model predicted as positive, the percentage that were actually positive): 0.6728, Recall(Of all the items that were actually positive, the percentage the model correctly identified): 0.9240 and test_fnc (FNC): Accuracy: 0.6296, F1-score: 0.4475, Precision: 0.3728, Recall: 0.5595.

To further evaluate these results, a ROC Curve(Receiver Operating Characteristic Curve) was used to plot the true positive rate against the false positive rate at varying classification thresholds. The AUC(Area Under the Curve) provides a value summarising the model's ability to discriminate between positive and negative classes. As presented in Figure 5.1, the AUC for the ByteDance test set is 0.98. This visually confirms the model's excellent capability to separate agreeing and disagreeing headline pairs in the test set and having a sharp curve that rises in the top-left, indicating the low false positive rates it achieves. However, in contrast to these excellent results, the ROC Curve of the FNC test set presented in Figure 5.2 only has an AUC of 0.64, signifying only a slight improvement over random guessing(AUC=0.5). The curve's shallower slope visually demonstrates the difficulty with which the model has distinguishing the headline-body pairs, leading to the low F1-score in the results.

Diving deeper into FNC's failure to generalise, one can see a significant performance drop between FNC's validation and test sets. The validation set achieved an F1-score of 0.55 in the best epoch, whereas the test set only yielded a much lower F1-score of 0.45 and a low AUC of 0.64 in the ROC Curve. Using a single-task RoBERTa model for news disagreement detection as a reference, it also struggled on the FNC test set with an even lower F1-score of 0.43. Therefore, due to the persistent difficulty observed across the multitask and single-task RoBERTa models, it strongly suggests issues beyond the learning paradigm. Potential causes for the difficulty in classifying FNC may be due to major distributional shifts between training/validation and test splits, higher complexity within FNC sets, which require more sophisticated reasoning than what is captured by RoBERTa alone or possible noise or inconsistencies in FNC's annotations within the test set itself.

To rigorously evaluate the final multitask RoBERTa model's performance for

disagreement detection, it is benchmarked against two informative baseline models: a multitask LSTM model representing a traditional recurrent architecture and a single-task RoBERTa model solely focusing on news disagreement detection. The F1-score results of the BD and FNC test sets will be used as a comparison to assess the benefit of transformer architecture (RoBERTa vs. LSTM) and the impact of a multitask model learning strategy versus a single-task model.

Table 5.2: Comparison of Disagreement Detection F1 Scores on Test Sets

Model Type	Test Set: BD F1	Test Set: FNC F1
Multitask RoBERTa	0.7786	0.4475
Single-Task RoBERTa	0.8719	0.4287
Multitask LSTM	0.7568	0.3529

Note: Best F1-score is shown in bold.

By assessing the results in Table 5.2, the multitask RoBERTa model versus the multitask LSTM model, it is shown that RoBERTa consistently outperformed the LSTM model across both test sets, which validates the choice of transformer architecture used for these complex NLP tasks. On the BD test set, RoBERTa achieved a modest improvement of 0.0218 over the F1 score of LSTM, and in the significantly more challenging FNC test set, the RoBERTa model achieved a substantially higher F1-score of 0.4475, which is 0.0946 more than LSTM’s 0.3529. Thus, we can conclude that the performance gap, particularly on the FNC test set, highlights the effectiveness of RoBERTa’s complex architecture compared to a more traditional architecture like LSTM. This highlights RoBERTa’s ability to understand more nuanced relationships and intricate linguistic patterns, which are essential for detecting disagreement, which means the more complex a task, the easier RoBERTa can handle it than LSTM. Even though the LSTM model offers faster training times, the substantial performance gains from RoBERTa justify the increased computational cost and complexity of this implementation.

Finally, a more detailed picture of the multitask learning approach is painted by comparing the single-task RoBERTa model to the final proposed multitask RoBERTa model. On the BD test set, the single-task model significantly outperforms the multitask model with a high F1-score of 0.8719 compared to the multitask model’s 0.7786. The 0.0933 difference in score suggests potential negative interference on the multitask approach for this dataset. Since the BD dataset may have a simpler or clearer representation of disagreement, when training simultaneously with the complex emotion recognition task, the noise of a different task could prevent the shared RoBERTa encoder from completely optimising its representations for the patterns within the BD test set. Therefore, the single-task model may

perform better because it is more effectively specialised to the task. However, the opposite results occurred on the FNC dataset, although within a smaller margin. The multitask model achieved an F1-score of 0.4475, slightly higher than the single task's F1-score of 0.4287. While both models performed poorly on the FNC test set, the multitask model saw an improvement of 0.0188 against the single-task model. This may suggest a potential minor positive interference from the emotion recognition task. It is plausible that the learning representations for emotion recognition introduced some form of regularisation or subtle linguistic cues that may have provided a marginal benefit for difficult FNC stance detection. Although due to the low F1-score and previously discussed generalisation issue with a low AUC of 0.64, the difference may also be within the range of experimental noise specific to the multitask model. In conclusion, the multitask model offered a marginal benefit to the performance on the difficult FNC data but appeared to perform significantly worse on the BD data, highlighting the common challenge in multitask learning, which is balancing task objectives and preventing negative interference, especially in tasks that differ significantly in complexity size. Additionally, the chosen strategy for the multitask model, selecting the best epoch with the minimum average validation loss across all tasks, may have favoured the model to perform moderately well overall rather than specifically excelling on disagreement datasets like BD.

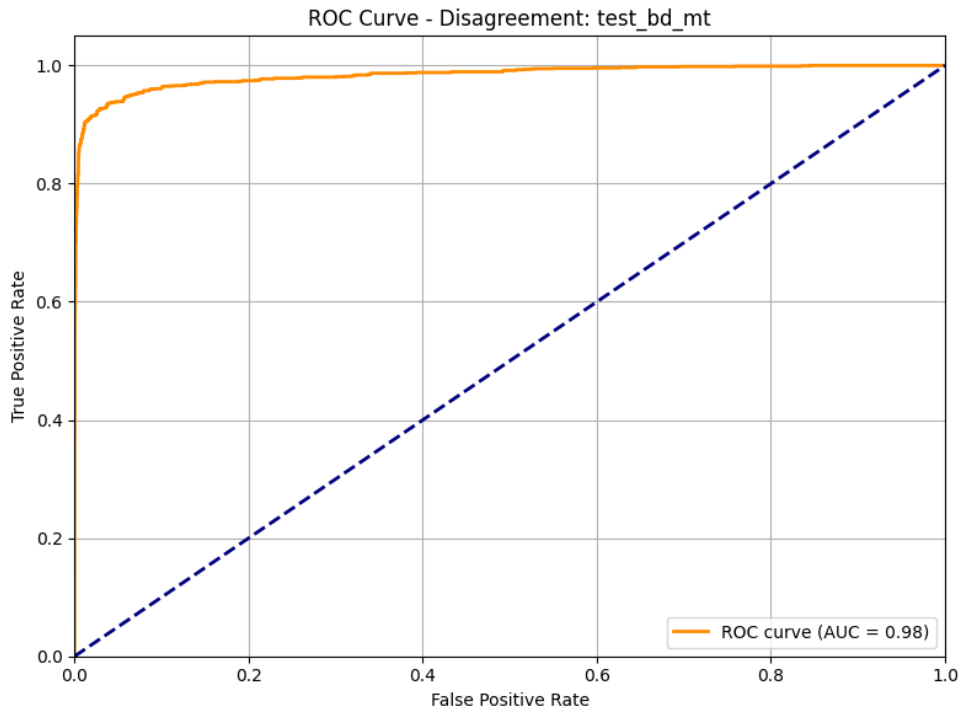


Figure 5.1: ROC Curve for Disagreement task on BD test set

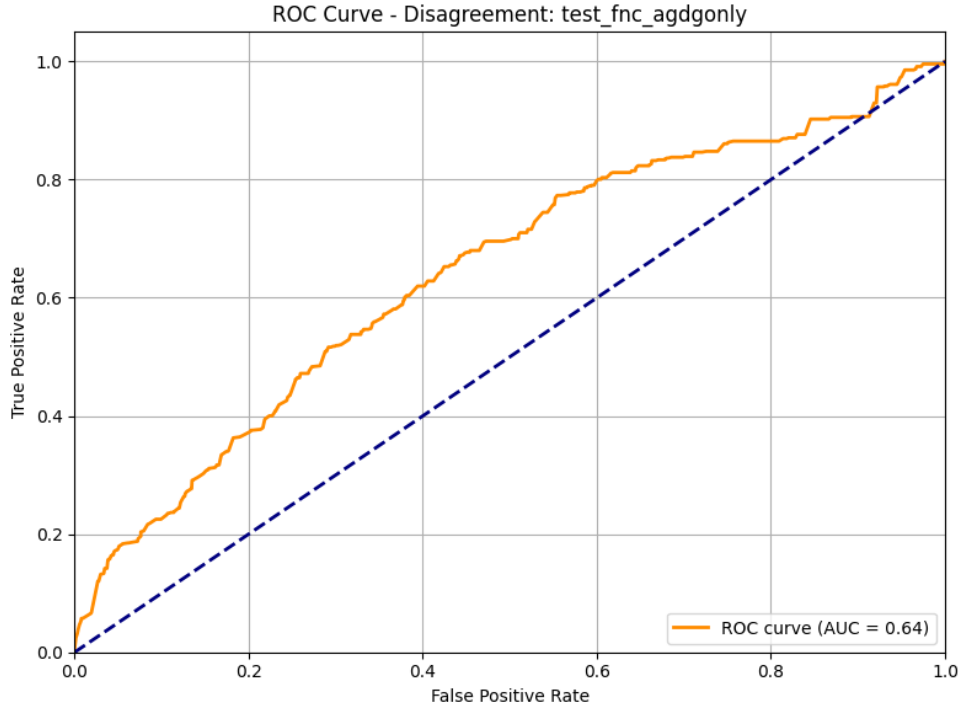


Figure 5.2: ROC Curve for Disagreement task on FNC test set

5.3 Performance Evaluation and Results - Emotion Recognition

The primary objective of this model was to focus on news disagreement detection and have an auxiliary task of emotion recognition. Therefore, the results for this task rely on the validation set results, as there were no dedicated test sets for the GoEmotions dataset. The best overall multitask RoBERTa model selected achieved the following performance metrics on the GoEmotions validation set: Accuracy: 0.2700, F1-score: 0.3093, Precision: 0.5159, and Recall: 0.3028. It is also noted that the F1-score did improve past this result. However, due to the best model being the average minimum validation loss, that model was not chosen because its loss was higher overall across all tasks.

Comparing the multitask RoBERTa model to the multitask LSTM, it is shown in Table 5.3 that the RoBERTa model performed marginally better than the LSTM model by scoring an F1-score of 0.3093, compared to 0.2939. These results contrast sharply with the disagreement task, where the RoBERTa model showed a clear advantage. Suggesting that the architectural benefits of a more complex transformer like RoBERTa are less pronounced for the emotion task within this specific instance of a challenging multitask approach or that the task itself is difficult for both architectures.

Table 5.3: Comparison of Emotion Recognition F1 Scores on GoEmotions Validation Set

Model Type	GoEmotions (Weighted) F1-score
Single-Task RoBERTa	0.4073
Multitask LSTM	0.2939
Multitask RoBERTa	0.3093

Note: Best F1-score is shown in bold.

Furthermore, when comparing the single-task RoBERTa model to the multitask RoBERTa model, the single-task model significantly outperformed the multitask model, scoring an F1-score of 0.4073, which is 0.098 higher than the multitask model’s 0.3093. Therefore, these results indicate performance degradation for the emotion recognition task when trained simultaneously with disagreement detection, as the multitask RoBERTa model barely outperforms the simpler multitask LSTM model on this specific task and performs considerably worse than its single-task model counterpart. A few possible hypotheses for this conclusion of performance degradation are task interference, where gradients from disagreement tasks potentially hinder optimisation for emotion nuances, differing convergence dynamics, which is due to disagreement tasks converging faster or stronger, thus leading the shared optimiser away from the optimum of the emotions task, and also representation dominance where disagreement tasks possibly dominate the fine-tuning of the shared encoder due to dataset size, which is further reinforced by the simpler multitask LSTM model also struggling with the emotion task, therefore showcasing the inherent difficulties in balancing these specific tasks.

In conclusion, evaluating the emotion recognition task reveals significant challenges in effectively training within the final proposed multitask RoBERTa model. While the RoBERTa architecture demonstrated clear advantages in news disagreement detection over the LSTM baseline model, the emotion recognition task only saw marginal improvements. Moreover, the final proposed multitask RoBERTa model performed significantly worse than its single-task counterpart, indicating negative interference when using a multitask approach with these specific tasks and potential performance degradation.

6 Conclusion

This project addressed the ever-growing issue of information disorder [2], by exploring the potential of Deep Multitask Learning (MTL) in enhancing automated news content analysis. The central hypothesis was that simultaneously training a model to detect factual disagreement and recognise the emotions conveyed could lead to more robust identification of misleading or false information [10], given that misinformation often leverages strong emotions to spread quicker [10].

To test this hypothesis, the final proposed multitask model went through the verification phase, which yielded mixed results regarding the efficacy of the MTL RoBERTa approach compared to the single-task baselines and a traditional LSTM model. The RoBERTa architecture [5] surpassed the LSTM baseline, particularly on the complex FNC dataset [7], which highlighted the advantages of transformers. However, compared to the single-task RoBERTa model, the MTL model's performance varied. The results showed that the MTL offers a slight edge on the FNC data but performed notably worse on the BD dataset [25], suggesting potential negative interference between tasks. Additionally, the multitask RoBERTa model significantly underperformed its single-task counterpart in the emotion recognition task and was only slightly better than the multitask LSTM. This indicates substantial reduction in performance for emotion detection within the MTL approach. This is likely caused by task interference or the disagreement task dominates the shared model components, reflecting the known challenges in multitask learning [11].

To conclude, the research successfully implemented a RoBERTa-based multitask model for news disagreement detection and emotion recognition. While validating the strength of transformer architectures [5] for these NLP tasks, there remain inherent issues of multitask learning. Balancing task objectives, reducing negative interference, and achieving strong generalisation across diverse datasets pose significant challenges. The strategies used in selecting the best model (by optimising for average validation loss across tasks) may have led to suboptimal performance on individual tasks. Future work should explore more advanced MTL techniques for better task balancing, methods to improve generalisation on complex data like FNC, along with incorporating complementary features beyond text content, such as social context [19], network dynamics [16][17], multi-modal data

6 Conclusion

[8][9] or evidence-based approaches [6]. Despite these challenges, this project contributes to the ongoing effort [1][3] to develop more sophisticated automated tools to combat the complex problem of information disorder online[2]

A Results of Each Model

```
Epoch 6 Average Training Loss (across 5 valid loaders): 0.5593

--- Validation ---
Evaluating on validation set: politifact
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 1.3597, Acc: 0.5871, F1: 0.3150, Prec: 0.5752, Rec: 0.2169
Evaluating on validation set: hyperpartisan
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 0.7394, Acc: 0.6512, F1: 0.5631, Prec: 0.5273, Rec: 0.6042
Evaluating on validation set: goemotions
Using 'weighted' averaging for metrics (max label = 27).
- Val Loss: 2.4196, Acc: 0.3028, F1: 0.3093, Prec: 0.5159, Rec: 0.3028
Evaluating on validation set: train_bd_mt
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 0.1827, Acc: 0.9915, F1: 0.9481, Prec: 0.9405, Rec: 0.9558
Evaluating on validation set: train_fnc_agdgonly
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 0.7483, Acc: 0.7489, F1: 0.5505, Prec: 0.4125, Rec: 0.8274

Epoch 6 Average Validation Loss (across 5 valid loaders): 1.0899
Validation loss improved (1.2004 --> 1.0899). Saving model to multitask_model_best.pth...
```

Figure A.1: The best validation epoch for the final proposed multitask model

```
--- Final Test Set Evaluation ---
Evaluating on Test Set: test_bd_mt
Using 'binary' averaging for metrics (max label = 1).
- Test Loss: 0.3087, Acc: 0.9695, F1: 0.7786, Prec: 0.6728, Rec: 0.9240
Evaluating on Test Set: test_fnc_agdgonly
Using 'binary' averaging for metrics (max label = 1).
- Test Loss: 1.2125, Acc: 0.6296, F1: 0.4475, Prec: 0.3728, Rec: 0.5595

--- Script Finished ---
```

Figure A.2: The test results of the final proposed multitask model

A Results of Each Model

```
Epoch 6 Average Training Loss: 1.0411

--- Validation ---
Evaluating on validation set: politifact
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 1.1744, Acc: 0.5759, F1: 0.3905, Prec: 0.5264, Rec: 0.3103
Evaluating on validation set: hyperpartisan
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 1.1701, Acc: 0.5349, F1: 0.3617, Prec: 0.3696, Rec: 0.3542
Evaluating on validation set: goemotions
Using 'weighted' averaging for metrics (max label = 27).
- Val Loss: 2.5335, Acc: 0.2571, F1: 0.2426, Prec: 0.3902, Rec: 0.2571
Evaluating on validation set: train_bd_mt
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 0.2598, Acc: 0.9763, F1: 0.8645, Prec: 0.8119, Rec: 0.9244
Evaluating on validation set: train_fnc_agdgonly
Using 'binary' averaging for metrics (max label = 1).
- Val Loss: 0.8629, Acc: 0.7566, F1: 0.5436, Prec: 0.4172, Rec: 0.7798

Epoch 6 Average Validation Loss: 1.2001
Validation loss improved (1.2444 --> 1.2001). Saving model to lstm_multitask_model_best.pth...
```

Figure A.3: The best validation epoch for the multitask LSTM baseline model

```
--- Final LSTM Test Set Evaluation ---
Evaluating LSTM on Test Set: test_bd_mt
Using 'binary' averaging for metrics (max label = 1).
- Test Loss: 0.3729, Acc: 0.9665, F1: 0.7568, Prec: 0.6551, Rec: 0.8959
Evaluating LSTM on Test Set: test_fnc_agdgonly
Using 'binary' averaging for metrics (max label = 1).
- Test Loss: 1.3762, Acc: 0.6912, F1: 0.3529, Prec: 0.4026, Rec: 0.3142

--- LSTM Script Finished ---
```

Figure A.4: The test results of the multitask LSTM baseline model

```
Epoch 5 Average Training Loss (across 4 valid loaders): 0.4618

--- Validation ---
Evaluating on validation set: politifact
Using 'binary' averaging for metrics.
- Val Loss: 1.3609, Acc: 0.6261, F1: 0.4009, Prec: 0.6710, Rec: 0.2859
Evaluating on validation set: hyperpartisan
Using 'binary' averaging for metrics.
- Val Loss: 0.7494, Acc: 0.7519, F1: 0.5789, Prec: 0.7857, Rec: 0.4583
Evaluating on validation set: train_bd_mt
Using 'binary' averaging for metrics.
- Val Loss: 0.1527, Acc: 0.9899, F1: 0.9396, Prec: 0.9161, Rec: 0.9643
Evaluating on validation set: train_fnc_agdgonly
Using 'binary' averaging for metrics.
- Val Loss: 0.7451, Acc: 0.7799, F1: 0.5702, Prec: 0.4475, Rec: 0.7857

Epoch 5 Average Validation Loss (across 4 valid loaders): 0.7520
Validation loss improved (0.8232 --> 0.7520). Saving model to disagreement_model_best.pth...
```

Figure A.5: The best validation epoch for the single-task disagreement baseline model

A Results of Each Model

```
--- Final Test Set Evaluation (Disagreement Task) ---  
Evaluating on Test Set: test_bd_mt  
Using 'binary' averaging for metrics.  
- Test Loss: 0.2955, Acc: 0.9840, F1: 0.8719, Prec: 0.8155, Rec: 0.9366  
Evaluating on Test Set: test_fnc_agdgonly  
Using 'binary' averaging for metrics.  
- Test Loss: 1.2384, Acc: 0.7150, F1: 0.4287, Prec: 0.4633, Rec: 0.3989  
  
--- Disagreement Script Finished ---
```

Figure A.6: The test results of the single-task disagreement baseline model

```
Epoch 4 Average Training Loss (across 1 valid loaders): 1.3923  
  
--- Validation ---  
Evaluating on validation set: goemotions  
Using 'weighted' averaging for metrics.  
- Val Loss: 1.6300, Acc: 0.4341, F1: 0.4073, Prec: 0.5527, Rec: 0.4341  
  
Epoch 4 Average Validation Loss (across 1 valid loaders): 1.6300  
Validation loss improved (1.6309 --> 1.6300). Saving model to emotion_model_best.pth...
```

Figure A.7: The best validation epoch for the single-task emotion baseline model

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